Workload Planning

Laboratory for the Information Systems course
in the degree program
Software Engineering and Media Computing

Milestone 3: System Design

submitted by

Team 4
Pavlos Arampatzis
Daniel Ryssel
Arbresha Selimi
Raluca Maria Vedislav

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Table of Contents

List of Figures

1 Local Data Models

Due to the view integration approach, the views were kept separate and are represented by separate data models. These models will get merged later, as described in the following Chapter 2, “Global Data Model.”

The data models were created in DbSchema. The views are realized as “Layouts”.

1.1 Subject Creation View

During the design the conclusion arose that the subject creation and service creation can be merged and that the data can be stored inside one “subject” table.

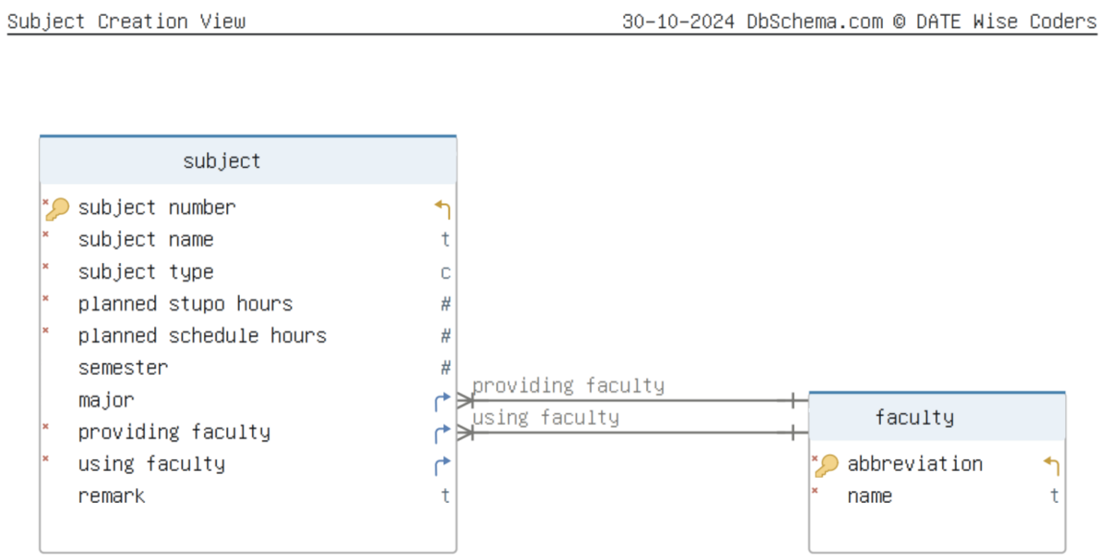

Figure 1: Subject Creation View

This layout, as seen in figure 1, shows the tables necessary for a subject and service creation.

1.2 Semester-Specific Course Planning View

The course planning for regular subjects and services was also merged into one table called “courses”.

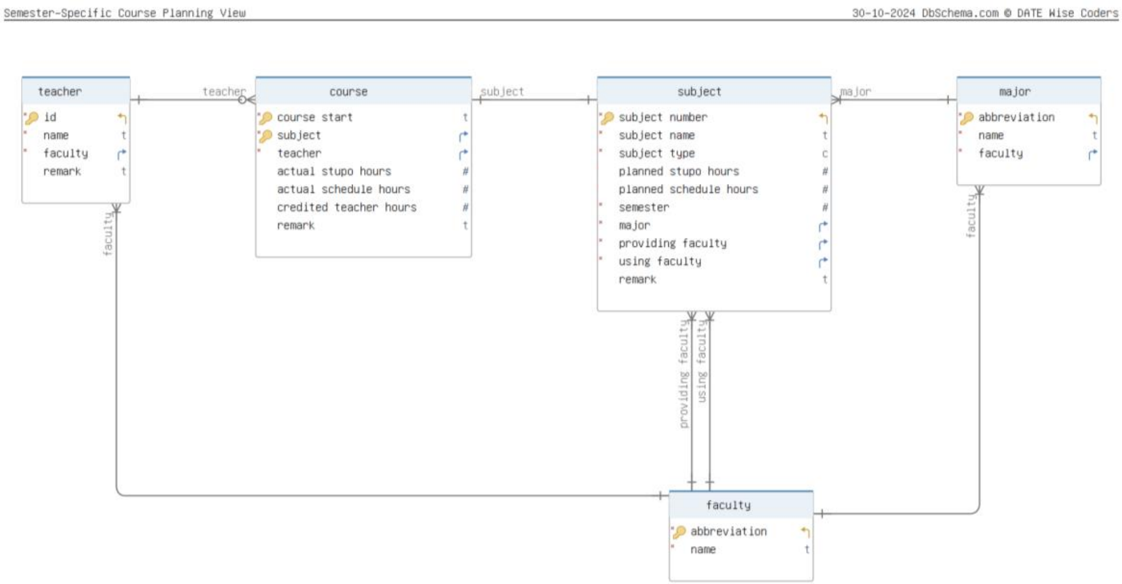

Figure 2: Semester-Specific Course Planning View

This layout, as seen in figure 2, shows the tables necessary for planning courses.

1.3 Workload Planning View

The workload planning view has been split into a view for a professor and a lecturer.

1.3.1 Professor Workload Planning View

Figure 3 on the following page shows the layout for the professor workload and contains all the necessary tables to manage the workload. The load reduction by performing functions is also considered.

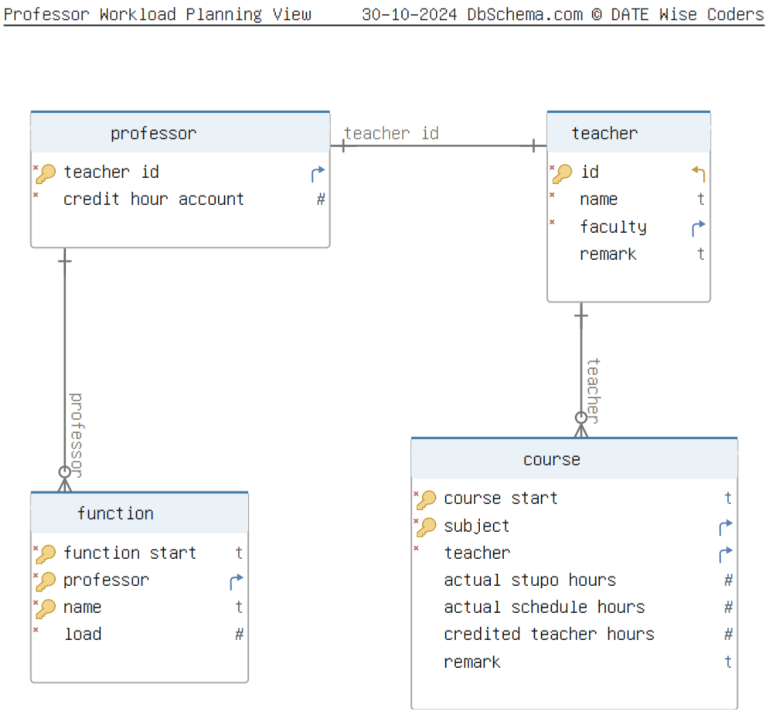

Figure 3: Professor Workload Planning View

1.3.2 Lecturer Workload Planning View

The following figure 4 shows the tables needed for this layout. In comparison to the professor, the lecturer cannot perform any functions and therefore does not have a credit hour account.

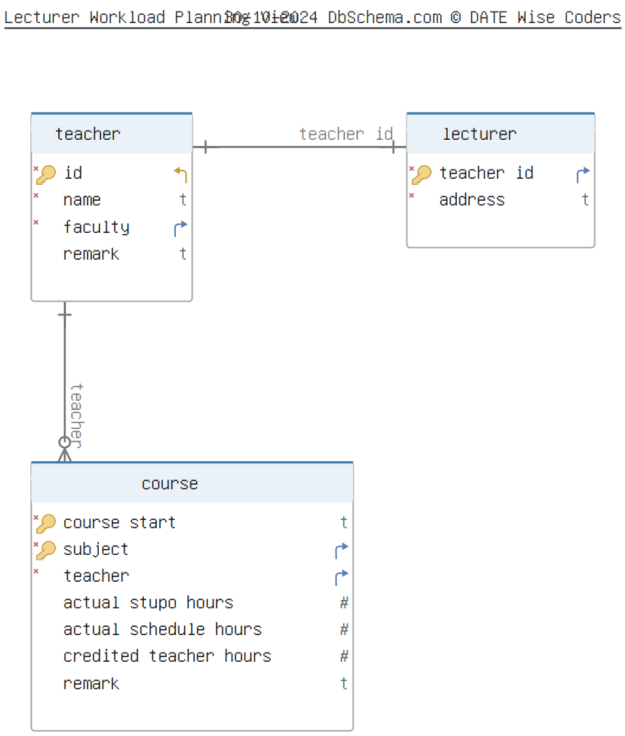

Figure 4: Lecturer Workload Planning View

1.4 Other

The other views that have been specified in Milestone 2 have been omitted. While working on the data models it turned out that some of them were not necessary. However, all information is covered.

2 Global Data Model

The following figure 5 shows the global data model and contains all local data models discussed before. Additional relationships had to be created in order to make sure that all data models are in relation to each other.

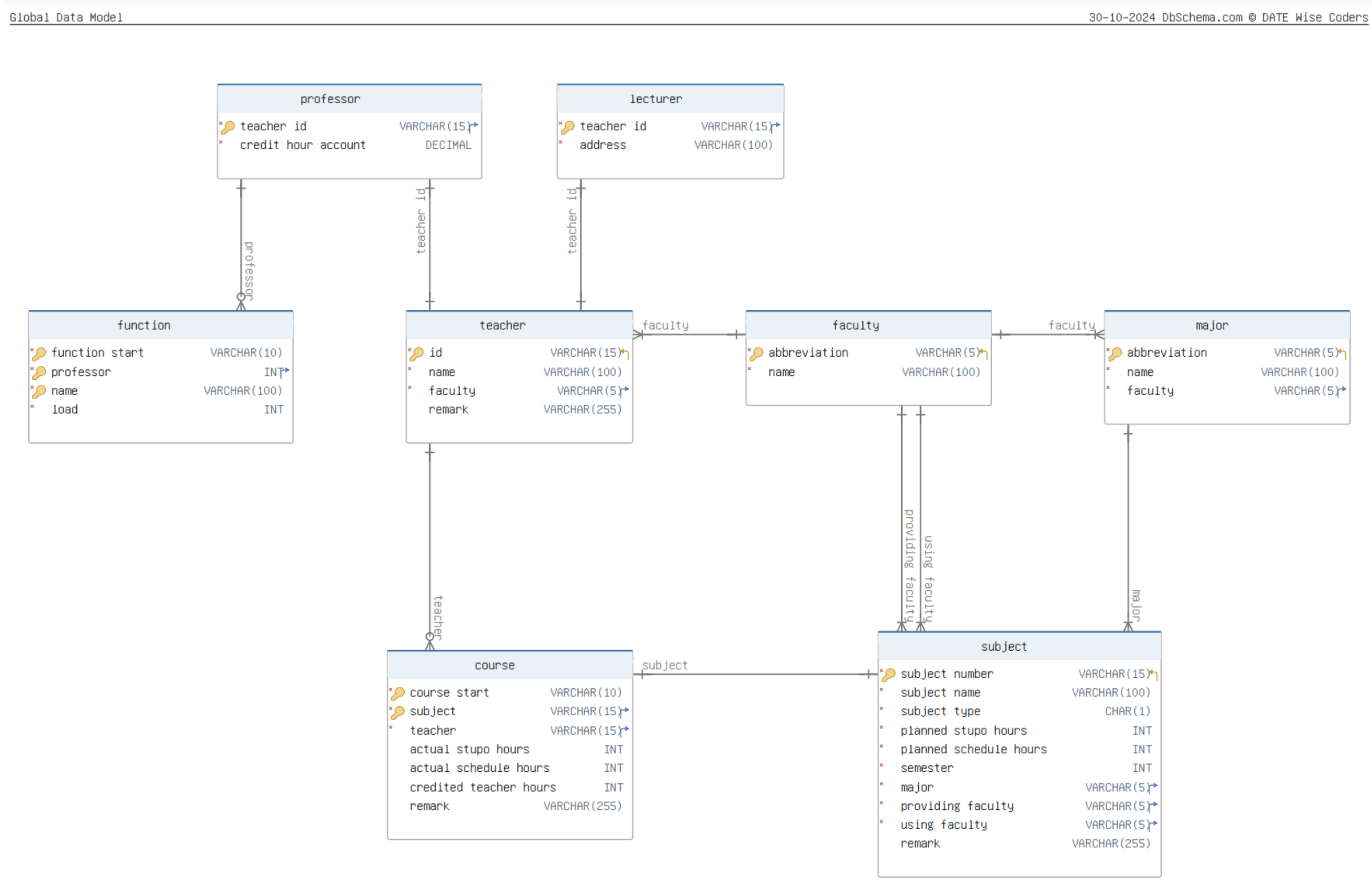

Figure 5: Global Data Model

3 Specifications

Lastly, the specifications were written for all tables and columns. Some include examples where necessary for clarification.

The specifications are formatted by DbSchema and are included in the pdf attachment after this page.

Global Data Model

Entity course

Idx	Name	Data Type	Description
A class that is held by a teacher in a specific semester (such as "WS24/25").			
* Pk	course start	VARCHAR(10)	The semester in which a class starts. E.g. "WS24/25"
* Pk	subject	VARCHAR(15)	The subject which is taught in this course. E.g. "Information Systems"
*	teacher	VARCHAR(15)	The teacher (id) teaching this course.
	actual stupo hours	INT	The actual hours for this subject according to the StuPo. E.g. "2"
	actual schedule hours	INT	The actual hours for this subject according to the schedule. E.g. "2"
	credited teacher hours	INT	The hours that will be credited to the teacher. This is part of the teacher's workload. E.g. "1"
	remark	VARCHAR(255)	An optional remark about the course.

Indexes

Type	Name	On	Description
Pk	pk_course	subject, course start	Primary key consisting of the course start and the subject number, which is a foreign key.

Relationships

Type	Name	On	Description
	fk_subject_teacher_subject (subject) ref subject (subject number)		
	fk_subject_teacher_teacher (teacher) ref teacher (id)		

Entity faculty

Idx	Name	Data Type	Description
An academic unit that specializes in a particular field and contains related subjects			
* Pk	abbreviation	VARCHAR(5)	An identifying abbreviation for the faculty. E.g. "IT"
*	name	VARCHAR(100)	The full name of the faculty. E.g. "Information Technology"

Indexes

Type	Name	On	Description
Pk	pk_faculty	abbreviation	Primary key consisting of the faculty abbreviation.

Entity function

Idx	Name	Data Type	Description
Responsibilities undertaken by a professor related to the management and organization of academic programs, faculty governance, and departmental operations.			
* Pk	function start	VARCHAR(10)	The semester in which a function starts. E.g. "WS24/25"
* Pk	professor	INT	The professor (id) which exercises this function.
* Pk	name	VARCHAR(100)	The name of the function. E.g. "Head of internship office"
*	load	INT	The hourly load of the function. E.g. "2"

Indexes

Type	Name	On	Description
Pk	pk_func_prof_name	function start, professor, name	Primary key consisting of the function start, the professor and the name of the function.

Relationships

Entity function

Type	Name	On	Description
	fk_function_professor (professor) ref professor (teacher id)		

Entity lecturer

Idx	Name	Data Type	Description
A contracted teacher, also known as "lecturer" in the project document.			
* Pk	teacher id	VARCHAR(15)	An identifying number for the teacher. Foreign Key from the teacher table.
*	address	VARCHAR(100)	Address of the lecturer. In order to be able to write to the lecturers. E.g. "Kanalstraße 33, Esslingen"

Indexes

Type	Name	On	Description
Pk	pk_lecturer	teacher id	Primary key consisting of the teacher id, which is also a foreign key.

Relationships

Type	Name	On	Description
	fk_lecturer_teacher (teacher id) ref teacher (id)		

Entity major

Idx	Name	Data Type	Description
A degree program.			
* Pk	abbreviation	VARCHAR(5)	An identifying abbreviation for the major. E.g. "SWB"
*	name	VARCHAR(100)	The full name for the major. E.g. "Software Engineering and Media Computing"
*	faculty	VARCHAR(5)	The faculty a major belongs to. E.g. "IT" for "Software Engineering and Media Computing"

Indexes

Type	Name	On	Description
Pk	pk_major	abbreviation	Primary key consisting of the major abbreviation.

Relationships

Type	Name	On	Description
	fk_major_faculty (faculty) ref faculty (abbreviation)		

Entity professor

Idx	Name	Data Type	Description
A permanently employed academic who is responsible for teaching and research, also known as "professor" in the project document.			
* Pk	teacher id	VARCHAR(15)	An identifying number for the professor. Foreign Key from the teacher table.
*	credit hour account	DECIMAL DEFAULT 0	Additional work beyond the 18 compulsory work hours for a professor is credited to a credit hour account, less work is debited from this account. E.g. "2"

Indexes

Type	Name	On	Description
Pk	pk_professor	teacher id	Primary key consisting of the professor id, which is also a foreign key.

Relationships

Type	Name	On	Description
	fk_professor_teacher (teacher id) ref teacher (id)		

Entity subject

Idx	Name	Data Type	Description
A specific subject listed in the curriculum of a major.			
* Pk	subject number	VARCHAR(15)	An identifying number for the subject. E.g. "R39.04995"
*	subject name	VARCHAR(100)	The name of a subject. E.g. "Information Systems"
*	subject type	CHAR(1) DEFAULT 'P'	If the subject is an compulsory "P", elective "W" or supplementary "Z" E.g. "P" for "Information Systems"
*	planned stupo hours	INT	The planned hours for this subject according to the StuPo. E.g. "2"
*	planned schedule hours	INT	The planned hours for this subject according to the schedule. E.g. "2"
*	semester	INT	The planned semester for this subject according to the curriculum. E.g. "6" for "Information Systems"
*	major	VARCHAR(5)	The major to which this subject belongs to. E.g. "SWB"
*	providing faculty	VARCHAR(5)	The faculty that is offering this subject. E.g. "IT"
*	using faculty	VARCHAR(5)	The faculty that is using this subject, in case of the subject being a service. If the subject is not a service, the providing faculty = using faculty. E.g. "IT"
	remark	VARCHAR(255)	An optional remark about the subject.

Indexes

Type	Name	On	Description
Pk	pk_subject	subject number	Primary key consisting of the subject number.

Relationships

Type	Name	On	Description
	fk_subject_faculty (providing faculty) ref faculty (abbreviation)		
	fk_subject_faculty_0 (using faculty) ref faculty (abbreviation)		
	fk_subject_major (major) ref major (abbreviation)		

Entity teacher

Idx	Name	Data Type	Description
A person teaching a course.			
* Pk	id	VARCHAR(15)	An identifying number for the teacher.
*	name	VARCHAR(100)	The name of the teacher. E.g. "Steffen Schober"
*	faculty	VARCHAR(5)	The faculty the teacher belongs to. E.g. "IT" for "Steffen Schober"
	remark	VARCHAR(255)	An optional remark made by the teacher about themselves.

Indexes

Type	Name	On	Description
Pk	pk_teacher	id	Primary key consisting of the teacher id.

Relationships

Type	Name	On	Description
	fk_teacher_faculty (faculty) ref faculty (abbreviation)		